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«УрФУ имени первого Президента России Б.Н. Ельцина» Кафедра
«Институт радиоэлектроники и информационных технологий-РТФ»**

**ПРАКТИЧЕСКАЯ РАБОТА №6.
HARDHAT ПРОЕКТ**

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Создание React приложения

```
PS D:\Codes\technology_rasp_reestr\5 lab\hardhat-tutorial> npm create vite@latest app --template react
◇ Starting dev server...
npm warn Unknown env config "template". This will stop working in the next major version of npm.

> app@0.0.0 dev
> vite

VITE v8.0.4 ready in 272 ms

→ Local:   http://localhost:5173/
→ Network: use --host to expose
→ press h + enter to show help
```

Инициализация Hardhat

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\Codes\technology_rasp_reestr\5 lab> npx hardhat --init
55 packages are looking for funding
  run `npm fund` for details

15 vulnerabilities (8 low, 5 moderate, 2 high)

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
✨ Dependencies installed ✨
Give Hardhat a star on Github if you're enjoying it! 🌟🌟

https://github.com/NomicFoundation/hardhat
PS D:\Codes\technology_rasp_reestr\5 lab>
```

Обновление hardhat.config.ts

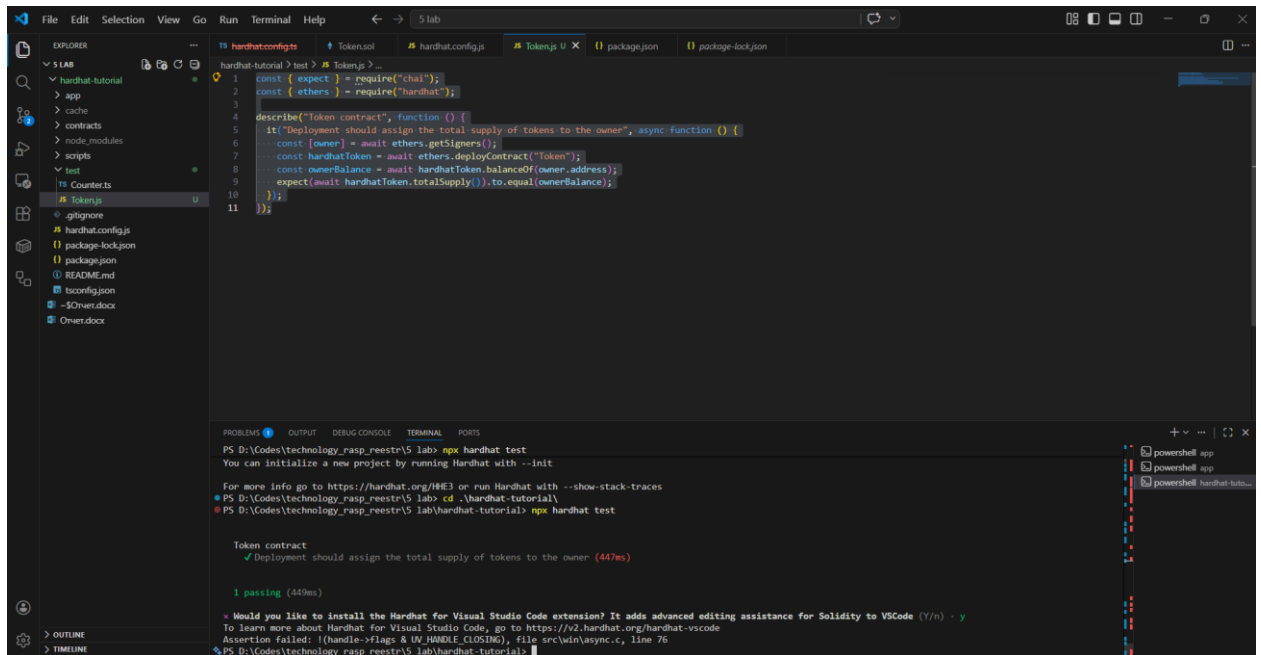
```
EXPLORER
5 LAB
  hardhat-tutorial
    > app
    > contracts
    > ignition
    > node_modules
    > scripts
    > test
    > .gitignore
    TS hardhat.config.ts U
    {} package-lock.json M
    {} package.json M
    @ README.md
    tsconfig.json U
    ~$Отчет.docx
    Отчет.docx

TS hardhat.config.ts U X
hardhat-tutorial > TS hardhat.config.ts >...
1  require("@nomicfoundation/hardhat-toolbox");
2
3  /** @type import('hardhat/config').HardhatUserConfig */
4  module.exports = {
5    solidity: "0.8.19",
6    paths: {
7      artifacts: "../app/src/artifacts",
8    },
9  };
```

Компиляция контракта

```
PS D:\Codes\technology_rasp_reestr\5 lab\hardhat-tutorial> npx hardhat compile
Downloading compiler 0.8.19
Compiled 1 Solidity file successfully (evm target: paris).
PS D:\Codes\technology_rasp_reestr\5 lab\hardhat-tutorial>
```

Запуск теста токена



```
hardhat-tutorial > test > Tokenjs > ...
1 const { expect } = require("chai");
2 const { ethers } = require("hardhat");
3
4 describe("Token contract", function () {
5   it("Deployment should assign the total supply of tokens to the owner", async function () {
6     const [owner] = await ethers.getSigners();
7     const hardhatToken = await ethers.deployContract("Token");
8     const ownerBalance = await hardhatToken.balanceOf(owner.address);
9     expect(await hardhatToken.totalSupply()).to.equal(ownerBalance);
10   });
11 });
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\Codes\technology_rasp_reestr\5 lab> npx hardhat test

You can initialize a new project by running Hardhat with --init

For more info go to <https://hardhat.org/HHE3> or run Hardhat with --show-stack-traces

PS D:\Codes\technology_rasp_reestr\5 lab> cd .\hardhat-tutorial\

PS D:\Codes\technology_rasp_reestr\5 lab\hardhat-tutorial> npx hardhat test

Token contract

✓ Deployment should assign the total supply of tokens to the owner (447ms)

1 passing (449ms)

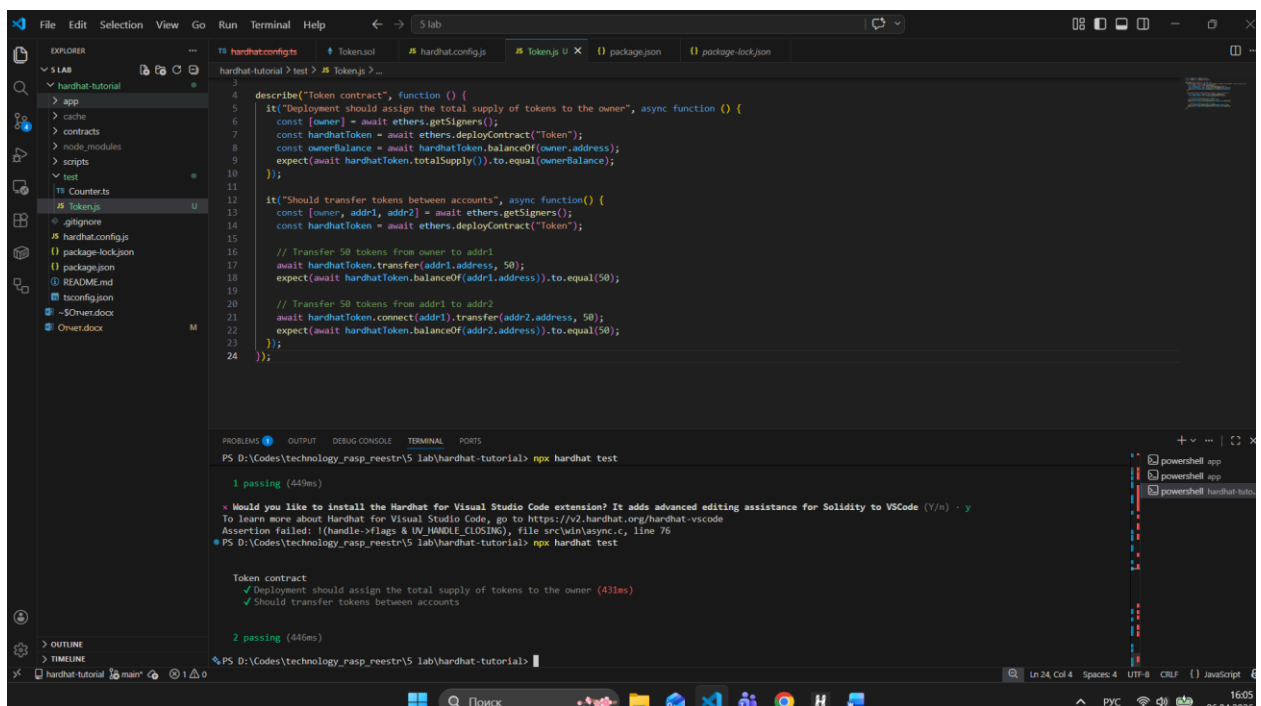
> Would you like to install the Hardhat for Visual Studio Code extension? It adds advanced editing assistance for Solidity to VSCode (Y/n) · y

To learn more about Hardhat for Visual Studio Code, go to <https://v2.hardhat.org/hardhat-vscode>

Assertion failed: ! (handle-yflags & UV_HANDLE_CLOSING), file src/win/async.c, line 76

PS D:\Codes\technology_rasp_reestr\5 lab\hardhat-tutorial>

И дополнение теста



```
hardhat-tutorial > test > Tokenjs > ...
3
4 describe("Token contract", function () {
5   it("Deployment should assign the total supply of tokens to the owner", async function () {
6     const [owner] = await ethers.getSigners();
7     const hardhatToken = await ethers.deployContract("Token");
8     const ownerBalance = await hardhatToken.balanceOf(owner.address);
9     expect(await hardhatToken.totalSupply()).to.equal(ownerBalance);
10   });
11
12   it("Should transfer tokens between accounts", async function() {
13     const [owner, addr1, addr2] = await ethers.getSigners();
14     const hardhatToken = await ethers.deployContract("Token");
15
16     // Transfer 50 tokens from owner to addr1
17     await hardhatToken.transfer(addr1.address, 50);
18     expect(await hardhatToken.balanceOf(addr1.address)).to.equal(50);
19
20     // Transfer 50 tokens from addr1 to addr2
21     await hardhatToken.connect(addr1).transfer(addr2.address, 50);
22     expect(await hardhatToken.balanceOf(addr2.address)).to.equal(50);
23   });
24 });
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\Codes\technology_rasp_reestr\5 lab\hardhat-tutorial> npx hardhat test

1 passing (449ms)

> Would you like to install the Hardhat for Visual Studio Code extension? It adds advanced editing assistance for Solidity to VSCode (Y/n) · y

To learn more about Hardhat for Visual Studio Code, go to <https://v2.hardhat.org/hardhat-vscode>

Assertion failed: ! (handle-yflags & UV_HANDLE_CLOSING), file src/win/async.c, line 76

PS D:\Codes\technology_rasp_reestr\5 lab\hardhat-tutorial> npx hardhat test

Token contract

✓ Deployment should assign the total supply of tokens to the owner (431ms)

✓ Should transfer tokens between accounts

2 passing (446ms)

PS D:\Codes\technology_rasp_reestr\5 lab\hardhat-tutorial>

Запуск Hardhat Network

```
PS D:\Codes\technology_rasp_reestr\5 lab\hardhat-tutorial> npx hardhat node

Account #16: 0x2546BcD3c84621a976D8185a91A922aE77ECe30 (10000 ETH)
Private Key: 0xea6c44ac03bff858b476bba40716402b03e41b8e97e276d1baec7c37d42484a0

Account #17: 0xbDA5747bFD65F08deb54cb465eB87D40e51B197E (10000 ETH)
Private Key: 0x689af8efa8c651a91ad287602527f3af2fe9f6501a7ac4b061667b5a93e037fd

Account #18: 0xd2FD4581271e230360230F9337D5c04308f44C0 (10000 ETH)
Private Key: 0xde9be858da4a475276426320d5e9262ecfc3ba460bfac56360bfa6c4c28b4ee0

Account #19: 0x8626f6940E2eb28930eFb4CeF49B2d1F2C9C1199 (10000 ETH)
Private Key: 0xdf57089febbacf7ba0bc227dafbffa9fc08a93fdc68e1e42411a14efcf23656e

WARNING: These accounts, and their private keys, are publicly known.
Any funds sent to them on Mainnet or any other live network WILL BE LOST.
```

Отправка запросов

```
PS D:\Codes\technology_rasp_reestr\5 lab\hardhat-tutorial\app> curl --data '{"jsonrpc":"2.0","method":"eth_getBalance","params":["0x8626f6940E2eb28930eFb4CeF49B2d1F2C9C1199"]}'
```

Деплой через Hardhat Ignition

```
1 const { buildModule } = require("@nomicfoundation/hardhat-ignition/modules");
2
3 module.exports = buildModule("TokenDeploy", (mbuilder) => {
4   const account0 = mbuilder.getAccount(0); // or account = "0x123..."
5   const erc20_smart_contract = mbuilder.contract("Token", [], { from: account0 });
6   return { erc20_smart_contract };
7 });

PS D:\Codes\technology_rasp_reestr\5 lab\hardhat-tutorial> npm install --save-dev @nomicfoundation/hardhat-ignition@0.15.16 @nomicfoundation/ignition-core@0.15.15
npm audit fix --force

Run "npm audit" for details.
PS D:\Codes\technology_rasp_reestr\5 lab\hardhat-tutorial> npx hardhat ignition deploy ignition/modules/TokenDeploy.js --network localhost
Hardhat Ignition

Deploying [ TokenDeploy ]

Batch #1
  Executed TokenDeploy#Token

[ TokenDeploy ] successfully deployed

Deployed Addresses

TokenDeploy#Token - 0x5FbDB2315678afecb367f032d93F642f64180aa3
```

TokenDeploy#Token - 0x5FbDB2315678afecb367f032d93F642f64180aa3

Настройка фронтенда

```
PS D:\Codes\technology_rasp_reestr\5 lab\hardhat-tutorial\app> npm i
npm audit fix --force

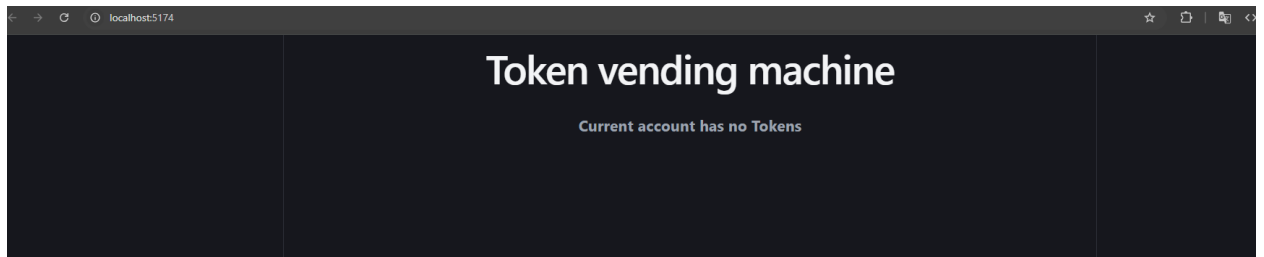
Run "npm audit" for details.
PS D:\Codes\technology_rasp_reestr\5 lab\hardhat-tutorial\app> npm run dev

> app@0.0.0 dev
> vite

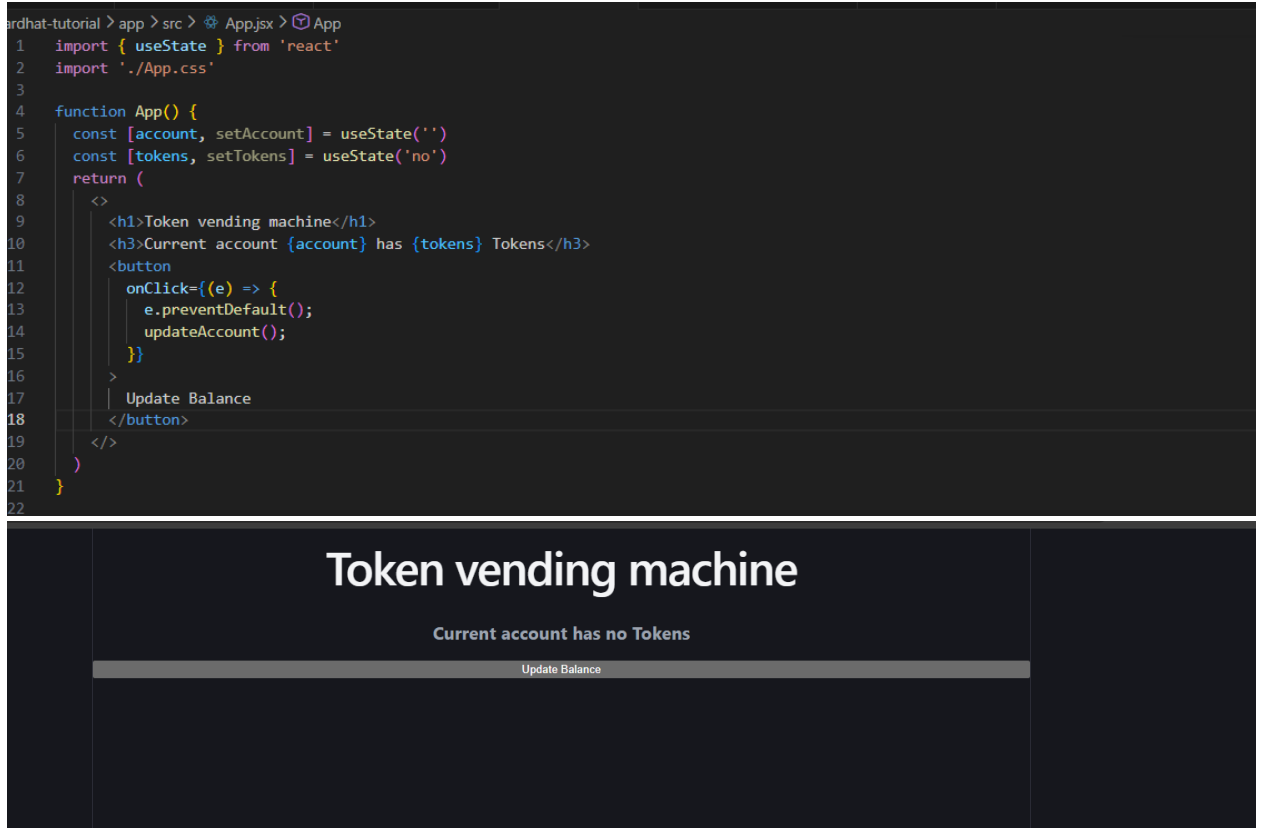
16:17:42 [vite] (client) Re-optimizing dependencies because lockfile has changed
Port 5173 is in use, trying another one...

VITE v8.0.4 ready in 197 ms

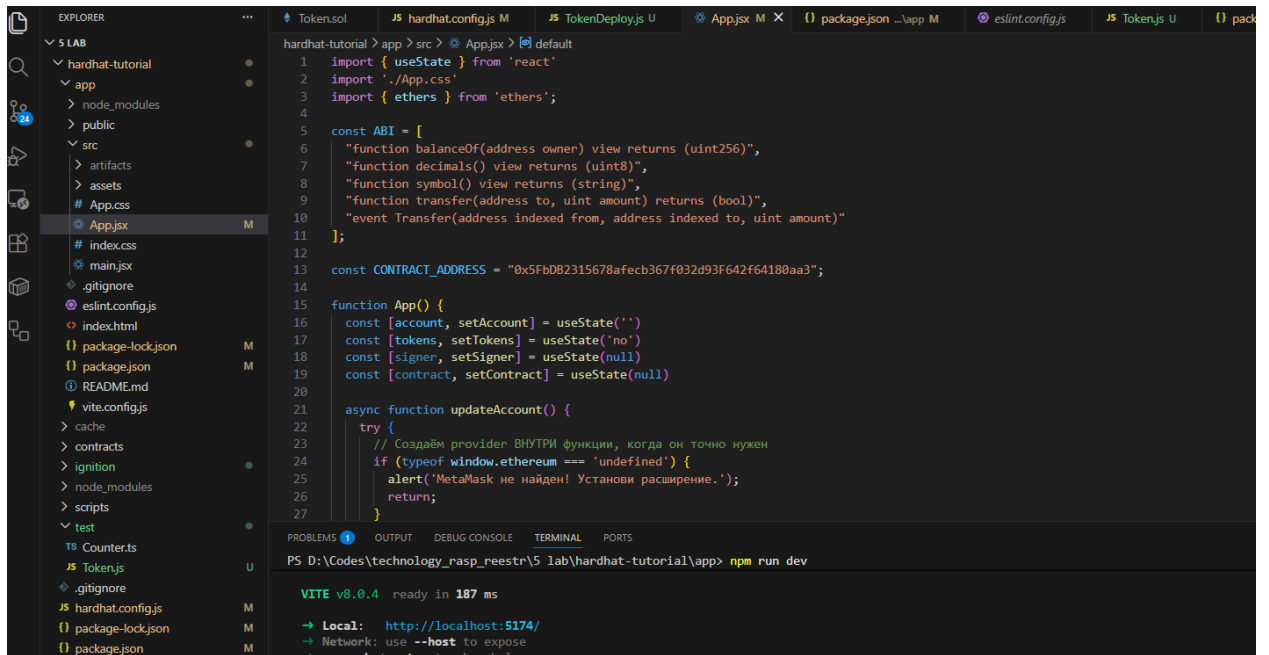
→ Local: http://localhost:5174/
→ Network: use --host to expose
→ press h + enter to show help
```



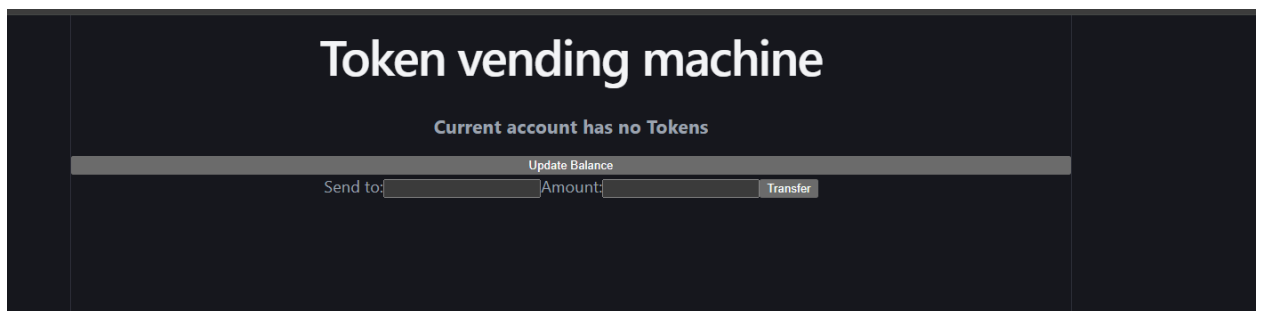
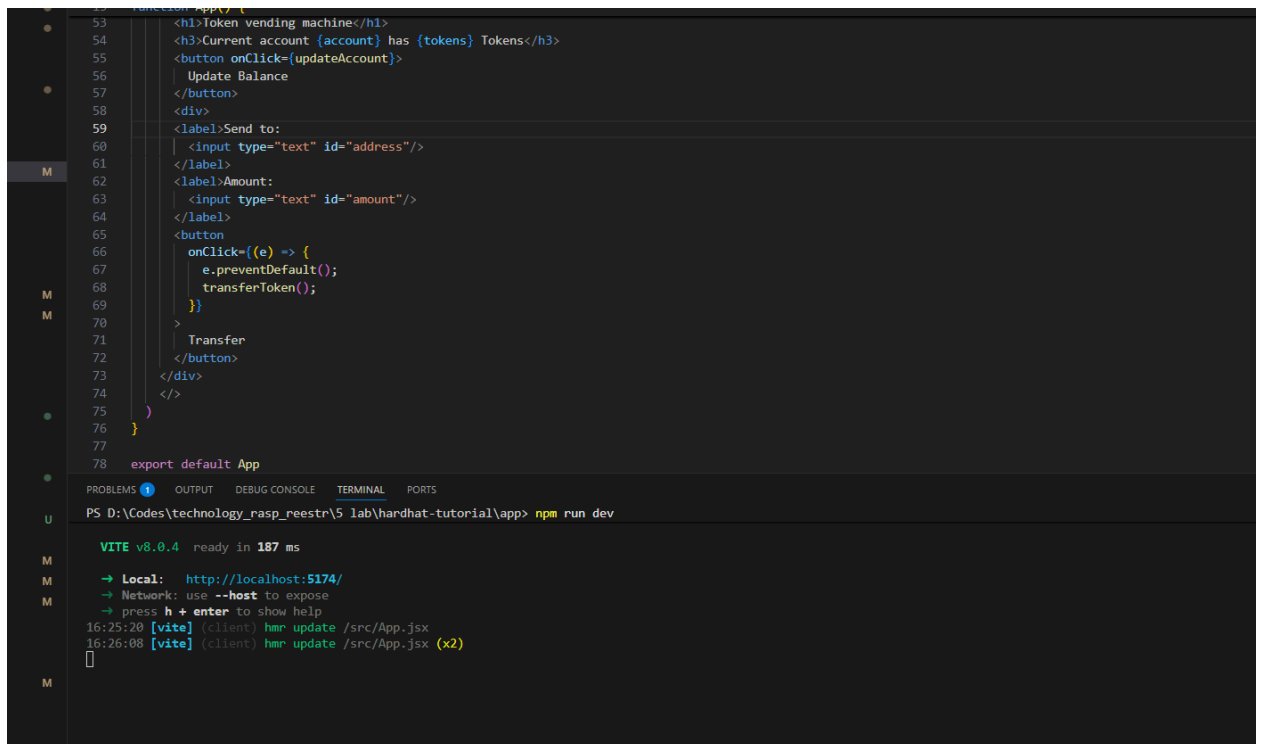
Добавление кнопки



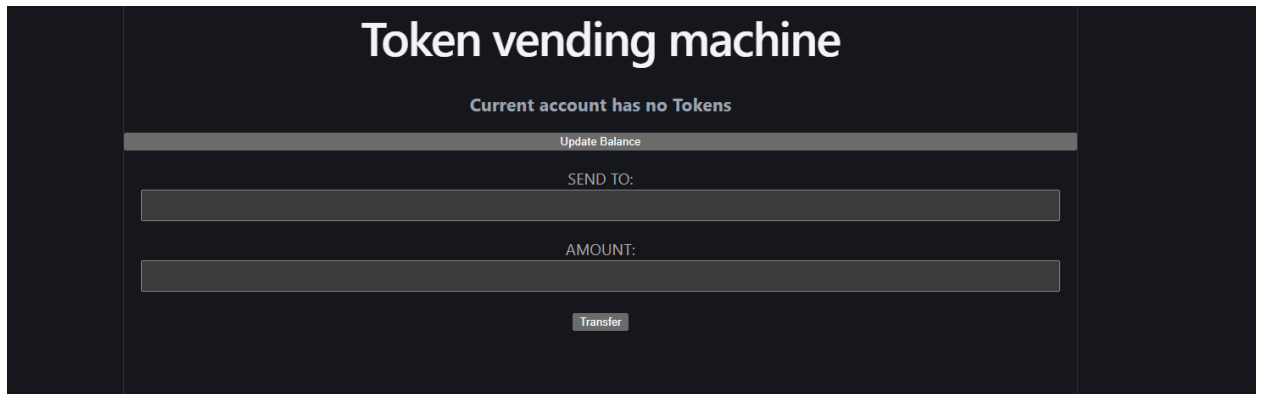
Подключение к смарт-контракту



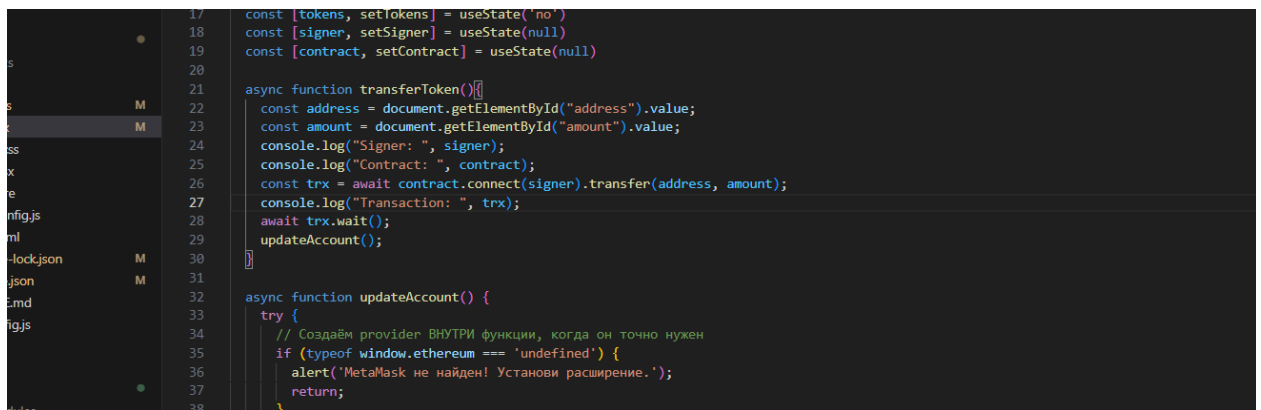
Интерфейс для перевода токенов



Обновление стилей



Добавление функции перевода токенов



Ссылка на github: https://github.com/MaksimKislyakov/technology_rasp_reestr